NIFTY 100

FINANCIAL INTELLIGENCE PLATFORM

Analyst Guide

A practical guide to company research, financial screening, peer analysis, valuation intelligence and API access.

Platform coverage	Current project test status
92 companies · 11 peer groups · 10 KPI peer metrics	143 automated tests passed
SQLite data store · Python analytics · FastAPI	Black formatting and Ruff lint checks passed

Prepared for analyst and project review use · Version 1.0 · 07 November 2026

Important: This platform is an analytical support tool, not investment advice. Validate figures against primary filings and apply sector-specific judgment before making investment decisions.

1. Platform Overview

The NIFTY 100 Financial Intelligence Platform consolidates company-level financial information and provides repeatable analytical workflows for screening, peer comparison, valuation review, portfolio context and report generation.

The project combines a SQLite database, Python-based ETL and analytics modules, a Streamlit dashboard, a FastAPI service layer and PDF tearsheets. The platform is designed to help analysts move from broad market coverage to focused company investigation.

Core capabilities

- **Financial data:** profit and loss, balance sheet, cash flow, market capitalisation and company metadata.
- **Computed KPIs:** profitability, returns, leverage, cash generation, conversion and growth metrics.
- **Screening:** six preset strategies to narrow the company universe.
- **Peer engine:** within-group percentile rankings across ten financial metrics.
- **Clustering:** KMeans grouping of 92 companies into five data-driven archetypes.
- **Valuation and reporting:** valuation summary outputs, portfolio statistics and company tearsheets.
- **API:** versioned REST endpoints for programmatic access and integration.

Coverage snapshot

Item	Known project coverage
Company master	92 companies
Peer groups	11 groups
Peer metrics	10 metrics
Cluster output	5 clusters; 92 company labels
Automated tests	143 passed; 2 dependency deprecation warnings

The company universe and record counts are project-state observations, not a guarantee that every company has complete history for every dataset.

2. Architecture and Data Flow

The platform follows a layered workflow. Source spreadsheets and prepared datasets are ingested, normalised and validated before being stored in SQLite. Analytics modules calculate ratios, screeners, peer ranks, clusters and valuation outputs. The dashboard and API expose those results to analysts and downstream consumers.

Layer	Responsibility	Main components
Source / ETL	Read source workbooks, normalise fields and periods, validate records	data_loader.py, normaliser.py, validator.py
Storage	Persist company, financial, market and analytical tables	data/nifty100.db
Analytics	Calculate KPIs, screeners, peers, clustering and valuation	src/analytics/
Presentation	Interactive analyst screens and downloadable outputs	Streamlit dashboard, reports/
Service	Expose data and analytics through REST endpoints	src/api/
Quality	Unit, integration and API regression tests	tests/

Typical data flow

Source workbooks → ETL loading → normalisation → validation → SQLite tables → KPI calculations → analytics outputs → Streamlit / FastAPI / PDF reports.

Database tables

The known database contains tables including companies, sectors, profitandloss, balancesheet, cashflow, financial_ratios, stock_prices, market_cap, peer_groups, documents, prosandcons and analysis. Exact column-level definitions should be taken from the live database schema or source code before external publication.

Operational note

The database file is expected at `data/nifty100.db`. Keep the project working directory consistent with the configured relative paths. Back up the database before rerunning ETL or any process that writes analytical tables.

3. Getting Started

Prerequisites

- Python 3.10 or a compatible project-supported version.
- A virtual environment with the project dependencies installed.
- The project repository and the SQLite database at data/nifty100.db.
- A terminal opened at the repository root.

Windows PowerShell setup

```
# Move to the project directory
cd "E:\Data Analytics\NIFTY 100 FINANCIAL\nifty100"

# Activate the virtual environment
.\venv\Scripts\Activate.ps1

# Install dependencies if requirements are maintained
pip install -r requirements.txt

# Run the automated test suite
pytest -v
```

Start the API

```
uvicorn src.api.main:app --host 127.0.0.1 --port 8000 --reload
```

Then open `http://127.0.0.1:8000/docs` to inspect the interactive OpenAPI documentation. Use the health endpoint to confirm that the API is responding and connected to its database.

Start the Streamlit dashboard

```
streamlit run src/dashboard/app.py
```

The dashboard is expected to run locally on port 8501 unless configured otherwise. If the command or entry-point differs in the checked-out branch, use the actual app entry point listed in the repository.

Do not run destructive ETL or database rebuild commands against the only copy of the database. Create a backup first.

4. Navigating the Dashboard

The Streamlit application is organised around analyst workflows. The intended dashboard navigation contains eight screens. Screen-specific labels and widgets may vary with the current implementation; use the sidebar navigation in the running application.

Screen	Analyst purpose	Suggested workflow
Home	Market-wide overview and key indicators	Start here; note coverage and broad distributions.
Company Profile	Company snapshot and financial history	Select a ticker; review sector, KPIs and historical statements.
Screener	Filter the company universe	Choose a preset; narrow by sector or supported thresholds.
Peers	Compare a company with its peer group	Review percentile ranks and relative strengths / weaknesses.
Trends	Explore historical movement	Inspect multi-period financial and KPI trends.
Sectors	Compare sector-level characteristics	Review company counts and sector-level medians.
Capital	Review cash flow and capital allocation signals	Interpret reinvestment, expansion, return and distress labels cautiously.
Reports	Access report and export workflows	Generate or retrieve tearsheets and analytical outputs.

Recommended analyst sequence

- Begin with Home to understand the available universe and data coverage.
- Use Screener to identify candidates; do not treat a preset match as a buy recommendation.
- Open the Company Profile and inspect the underlying statements and multi-year ratios.
- Use Peers and Sectors to compare like-with-like businesses.
- Review Trends, Capital and valuation indicators before documenting a conclusion.
- Export the relevant report and record any missing data or unusual values.

A dashboard chart is only as reliable as its underlying period coverage and data definitions. Check the reporting period and units before comparing companies.

5. Using the Financial Screener

The screener evaluates the latest available financial-ratio record for each company and applies a selected preset. The six documented presets are quality, low_debt, cash_generators, high_returns, efficient and balanced.

Preset	Documented screening logic
quality	ROE ≥ 15%; ROCE ≥ 15%; net profit margin ≥ 10%
low_debt	Debt-to-equity ≤ 0.5; interest coverage ≥ 3
cash_generators	Free cash flow > 0; CFO quality score ≥ 1
high_returns	ROE ≥ 20%; ROCE ≥ 20%
efficient	Asset turnover ≥ 0.75; operating profit margin ≥ 15%
balanced	Net profit margin ≥ 10%; ROE ≥ 12%; debt-to-equity ≤ 1; FCF > 0

Practical workflow

- Select one preset from the screener control.
- Optionally filter by broad sector if the dashboard/API supports it.
- Review the returned company count and ranking order.
- Inspect the actual metric values for each result; thresholds are filters, not a composite investment score.
- Export the results to CSV where the dashboard provides the download control.
- Open each shortlisted company profile and verify the latest reporting period.

Interpretation cautions

A company may be excluded because one required metric is missing, even if its business quality is strong. The quality preset is a rule-based filter, not a credit rating or recommendation. Financial companies may require different leverage and cash-flow interpretations from non-financial companies.

The documented Python screener sorts matching companies by ROE descending after applying the preset. Confirm the current API response and dashboard implementation if ranking behaviour is material to an analysis.

6. Company Profiles, KPIs and Peer Analysis

Company profile review

For a selected ticker, review the company identity, broad and sub-sector classification, latest available KPIs and financial statement history. The API exposes dedicated routes for profit and loss, balance sheet, cash flow and ratios. Use the same period convention when comparing statements.

Core KPI families

Family	Examples	Analyst question
Profitability	Net profit margin, operating profit margin	How much profit is generated from revenue?
Returns	ROE, ROCE	How efficiently is capital being used?
Leverage	Debt-to-equity, interest coverage	Can the company service its obligations?
Efficiency	Asset turnover	How effectively are assets converted into sales?
Cash flow	Free cash flow, CFO quality, FCF conversion	Are reported earnings supported by cash?
Growth	Revenue CAGR, PAT CAGR, EPS CAGR	What is the direction and pace of growth?

Peer analysis

The peer engine calculates within-peer-group percentile ranks for ten metrics: net profit margin, operating profit margin, ROE, ROCE, debt-to-equity, interest coverage, asset turnover, free cash flow, CFO quality score and FCF conversion. It also calculates an overall percentile/rank from the metric percentile columns.

For debt-to-equity, lower values are treated as better; the other documented metrics are ranked with higher values treated as better. This is a mechanical ranking convention and should not replace sector-specific judgement.

Peer groups contain a benchmark designation. Confirm the group composition and benchmark company before drawing conclusions from percentile positions.

7. Clustering, Trends and Sector Context

Clustering method

The project's company archetype workflow uses KMeans to assign 92 companies to five clusters. The documented feature set is ROE, debt-to-equity, five-year revenue CAGR, five-year free-cash-flow CAGR and operating profit margin. Features are standardised before clustering; missing feature values are imputed using sector medians. The configured random state is 42.

How to use cluster outputs

- Treat cluster labels as descriptive archetypes, not quality grades.
- Review cluster profiles and feature distributions before interpreting a group.
- Inspect companies in very small clusters individually; small groups may be driven by unusual values or missing-data patterns.
- Use the outlier report and correlation heatmap as diagnostic aids, not automatic exclusion rules.

Outliers and correlation

The project computes sector-relative Z-scores and flags values whose absolute Z-score exceeds 3. The outlier report contains flagged metric observations; the number of rows should not be interpreted as the number of distinct companies. Correlation analysis summarises relationships among selected KPIs across the company universe.

Sector context

Sector summaries support broad comparison through company counts and median indicators such as ROE, P/E and debt-to-equity where available. Sector medians are useful context but can hide wide variation among companies.

Caution: BEL, HAL and INDIGO were previously identified as extreme ROE outliers. Preserve and investigate these observations rather than automatically capping or deleting them.

8. Cash Flow, Capital Allocation and Valuation

Cash-flow intelligence

Cash-flow analysis combines operating cash flow, free cash flow and related conversion/quality indicators. The project's prior run classified CFO quality into High, Moderate, Accrual Risk and Insufficient Data. These are heuristic categories intended to prioritise review.

Capital allocation labels

The capital allocation workflow classifies observed patterns into labels such as Reinvest, Expansion, Return, Distress and Neutral. Labels are based on available annual financial records and should be read as indicators of observed patterns, not definitive management-intent assessments.

Known methodology caveats

- CapEx intensity has been approximated using the absolute value of net investing cash flow divided by sales; this is not a pure capital-expenditure measure.
- Missing cash-flow records must remain missing; do not substitute zero.
- ATGL was previously noted as having no cash-flow data in the project dataset.
- Banks, insurers and other financial institutions require different interpretation of cash-flow and leverage ratios.

Valuation review

The valuation workflow produces a valuation summary with indicators including FCF yield, P/E flags and overvaluation/discount labels. Historical market-cap data also supports review of valuation multiples such as P/E, P/B, EV/EBITDA and dividend yield where records exist.

A valuation flag is a screening signal, not an intrinsic-value conclusion. Confirm market price date, earnings period, share count assumptions and sector comparability before using it in a decision.

9. PDF Tearsheets and Reports

The project has generated company-level PDF tearsheets and sector-level reports. The known Sprint 5 output comprised 92 company tearsheets, each designed as a two-page report, along with ten sector PDFs and a portfolio summary.

Typical report workflow

- Open the Reports screen or use the company tearsheet API route.
- Select the company ticker and retrieve the corresponding PDF.
- Check the report date, financial period and data availability notes.
- Review the profile, key financial indicators, trends and any flags.
- Record any missing statements, unusual ratios or peer-group limitations.
- Retain the report with the analysis date for auditability.

API tearsheet route

```
GET /api/v1/companies/{ticker}/tearsheet

Example:
curl -L "http://127.0.0.1:8000/api/v1/companies/TCS/tearsheet" ^
  -o TCS_tearsheet.pdf
```

Report quality checks

Before distribution, confirm that the PDF opens, the company/ticker is correct, the period labels are visible, tables do not overflow, and missing values are clearly represented. A generated report should not imply a value exists where the underlying database has no record.

Known output location

Company PDFs were generated under `reports/tearsheets/`. Other reports and analytical artefacts are stored under `reports/` and `output/`.

10. REST API Reference

The API is built with FastAPI and uses the version prefix `/api/v1`. The running API's interactive documentation at `/docs` is the authoritative reference for the deployed code.

Method	Route	Purpose
GET	/api/v1/health	Service status and database health information
GET	/api/v1/companies	Company list with supported filters / search
GET	/api/v1/companies/{ticker}	Company profile
GET	/api/v1/companies/{ticker}/pl	Profit and loss history
GET	/api/v1/companies/{ticker}/bs	Balance sheet history
GET	/api/v1/companies/{ticker}/cashflow	Cash-flow history
GET	/api/v1/companies/{ticker}/ratios	Computed KPI history
GET	/api/v1/companies/{ticker}/tearsheet	Download company PDF
GET	/api/v1/screener	Run one of six screener presets
GET	/api/v1/sectors	Sector summary
GET	/api/v1/sectors/{sector}/companies	Companies in a sector
GET	/api/v1/peers/{group_name}	Peer-group rankings
GET	/api/v1/companies/{ticker}/peers/compare	Company-to-peer comparison
GET	/api/v1/market-cap/{ticker}	Historical valuation multiples
GET	/api/v1/portfolio/stats	Portfolio KPI distribution statistics
GET	/api/v1/companies/{ticker}/documents	Company annual-report document links
GET	/api/v1/valuation/summary	Valuation summary by requested year

This route list reflects the documented 18-path OpenAPI inventory from the project sprint notes. Always confirm the current OpenAPI schema before building a client against it.

11. API Examples

Health check

```
curl "http://127.0.0.1:8000/api/v1/health"
```

List companies and search by ticker

```
curl "http://127.0.0.1:8000/api/v1/companies"
curl "http://127.0.0.1:8000/api/v1/companies?search=TCS"
```

Retrieve a company profile and financial ratios

```
curl "http://127.0.0.1:8000/api/v1/companies/TCS"
curl "http://127.0.0.1:8000/api/v1/companies/TCS/ratios"
curl "http://127.0.0.1:8000/api/v1/companies/TCS/pl"
curl "http://127.0.0.1:8000/api/v1/companies/TCS/bs"
curl "http://127.0.0.1:8000/api/v1/companies/TCS/cashflow"
```

Run a screener

```
curl "http://127.0.0.1:8000/api/v1/screener?preset=quality"
curl "http://127.0.0.1:8000/api/v1/screener?preset=low_debt&sector=IT"
```

Peer group, valuation and portfolio statistics

```
curl "http://127.0.0.1:8000/api/v1/peers/IT"
curl "http://127.0.0.1:8000/api/v1/valuation/summary?year=2024"
curl "http://127.0.0.1:8000/api/v1/portfolio/stats"
```

The peer group name and supported query parameters depend on the database and current route implementation. A 404 indicates an unknown ticker, sector, peer group or missing report resource; a 422/400 response indicates invalid input according to the endpoint contract.

12. Data Quality, Limitations and Interpretation

Analysts must distinguish a true zero from an unavailable value. A missing statement, missing period or failed ticker match should remain explicitly missing unless a documented business rule supports a transformation.

Known data and acceptance gaps

Area	Known state / interpretation
Company master	92 companies in the companies table.
Financial ratios	1,073 records in the known database snapshot; project target is at least 1,100.
Pros / cons	521 generated signals: 432 pros and 89 cons. 90 companies had at least one pro, 52 at least one con, and 50 had both.
Analysis narratives	Only a small subset had matched analysis records in the previous check; do not infer coverage for all companies.
Cash flow	ATGL had no cash-flow data in the known snapshot; missing data is not zero.
Outliers	Extreme ROE observations were noted for BEL, HAL and INDIGO; preserve and investigate.

Important analytical limitations

- CAGR is sensitive to start/end values, period selection and negative-to-positive or positive-to-negative transitions.
- ROE can become extreme or misleading when equity is small or negative.
- Debt-to-equity and cash-flow ratios are not directly comparable across all financial sectors.
- Peer percentiles depend on peer-group composition and the availability of values for each metric.
- KMeans clusters depend on feature selection, scaling, imputation and the selected number of clusters.
- Heuristic pros/cons, distress, CFO quality and capital allocation labels are prioritisation aids, not calibrated probabilities.

Never alter the data merely to make an acceptance gate pass. Record the gap, explain its impact and propose a traceable remediation.

13. Troubleshooting and Support

Symptom	Checks / resolution
Database not found	Confirm current directory is the repository root and data/nifty100.db exists.
API fails to start	Activate the venv, install dependencies, check import errors and confirm port 8000 is free.
Dashboard fails to start	Run the documented Streamlit entry point and check dependency versions and port 8501.
Ticker returns 404	Check ticker spelling and whether the company exists in the companies table.
Peer group returns 404	Use an existing peer group name from peer_groups; check case and spaces.
Empty screener result	Review preset thresholds, missing latest KPI values and optional sector filter.
Unexpected null values	Check source coverage and ETL validation; do not replace missing values with zero.
PDF missing	Confirm reports/tearsheets contains the expected ticker PDF and that report generation completed.
Tests fail after changes	Run the failing test in isolation, inspect the traceback, then rerun the complete suite.

Quality commands

```
python -m black src tests scripts
python -m ruff check src tests scripts
pytest -v
```

Latest recorded code-quality result

The Day 44 check recorded Black formatting success, Ruff “All checks passed”, and 143 pytest tests passed with two dependency deprecation warnings. Re-run these commands after any further code change; these results are not a guarantee about later revisions.

14. Analyst Workflow and Decision Checklist

Suggested end-to-end research workflow

- **Define the question:** specify whether the task is screening, company review, peer comparison, valuation or portfolio context.
- **Check coverage:** verify company identity, latest financial year and statement availability.
- **Screen:** select an appropriate preset and record filters and result count.
- **Review fundamentals:** inspect P&L;, balance sheet, cash flow and KPI history.
- **Compare peers:** verify peer group fit, benchmark and percentile definitions.
- **Assess risk:** review leverage, cash conversion, distress signals and outlier flags.
- **Assess valuation:** inspect multiples and FCF yield with the correct date and period context.
- **Document:** generate a tearsheet, record caveats and preserve the analysis date.

Pre-publication checklist

- Ticker and company name match.
- Reporting period and currency/unit labels are clear.
- Missing values are identified and not silently converted to zero.
- Sector and peer group are appropriate.
- Extreme ratios have been investigated.
- Heuristic labels are described as signals, not definitive judgments.
- Any data gaps or failed validation checks are disclosed.
- The report is reproducible from the stated database snapshot and code version.

Use of outputs

The platform is intended to support research workflows and improve consistency. Analysts remain responsible for checking primary source documents, applying accounting and sector expertise, and documenting assumptions. The platform does not provide personalised financial advice or guarantee investment outcomes.

End of Analyst Guide

NIFTY 100 Financial Intelligence Platform · 07 November 2026